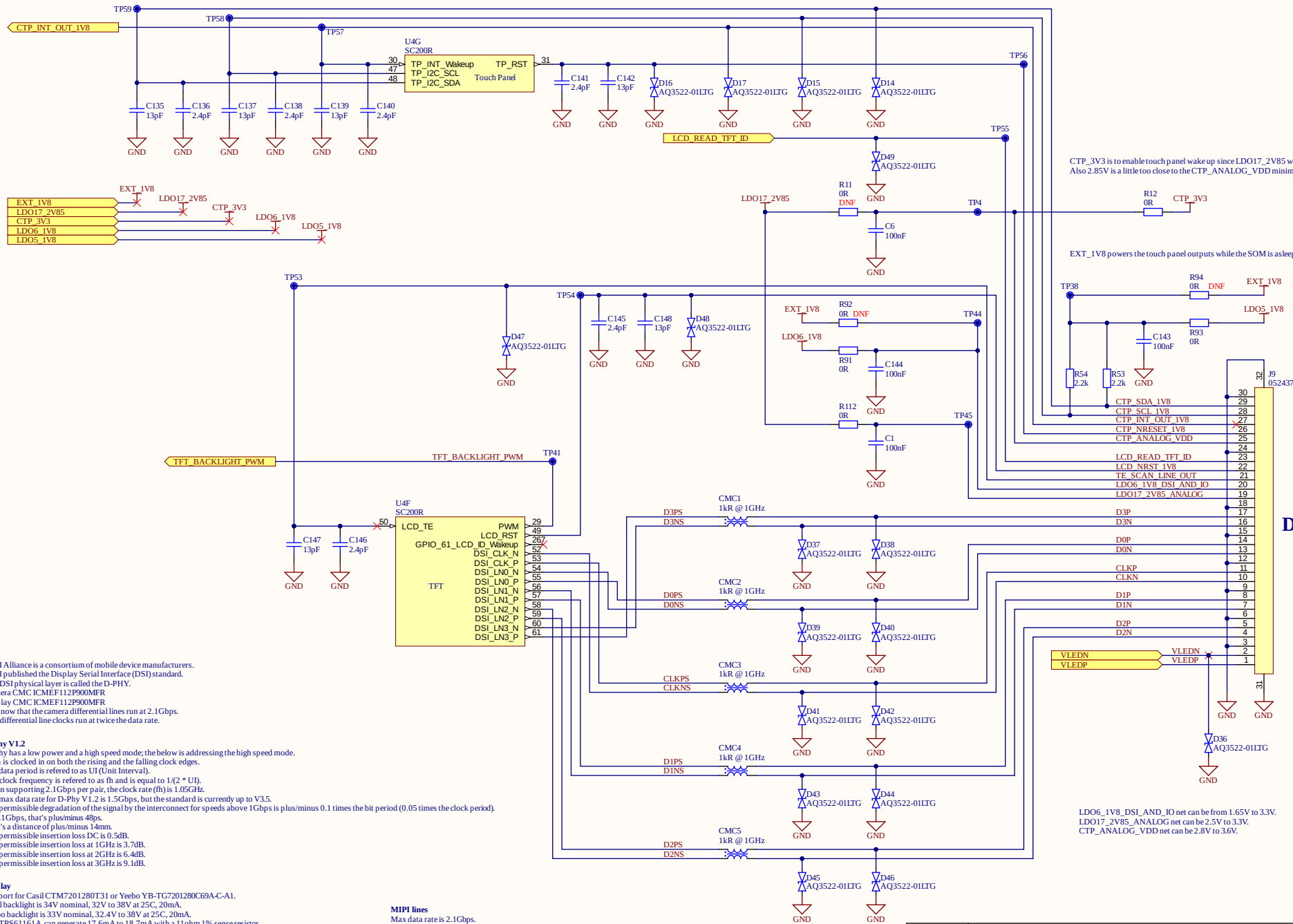


CTP_3V3 is to enable touch panel wake up since LDO17_2V85 will be off when SOM is asleep.
Also 2.85V is a little too close to the CTP_ANALOG_VDD minimum of 2.8V.

EXT_1V8 powers the touch panel outputs while the SOM is asleep and powers the STM8.

Display Connector

MIPI Alliance is a consortium of mobile device manufacturers.
MIPI published the Display Serial Interface (DSI) standard.
The DSI physical layer is called the D-PHY.
Camera CMC ICMEF112P900MFR
Display CMC ICMEF112P900MFR
We know that the camera differential lines run at 2.1Gbps.
The differential line clocks run at twice the data rate.

D-Phy V1.2
D-Phy has a low power and a high speed mode; the below is addressing the high speed mode.
Data is clocked in on both the rising and the falling clock edges.
The data period is refered to as UI (Unit Interval).
The clock frequency is refered to as fh and is equal to 1/(2 * UI).
When supporting 2.1Gbps per pair, the clock rate (fh) is 1.05GHz.
The max data rate for D-Phy V1.2 is 1.5Gbps, but the standard is currently up to V3.5.
The permissible degradation of the signal by the interconnect for speeds above 1Gbps is plus/minus 0.1 times the bit period (0.05 times the clock period).
At 2.1Gbps, that's plus/minus 48ps.
That's a distance of plus/minus 14mm.
The permissible insertion loss DC is 0.5dB.
The permissible insertion loss at 1GHz is 3.7dB.
The permissible insertion loss at 2GHz is 6.4dB.
The permissible insertion loss at 3GHz is 9.1dB.

Display
Support for Casil CTM7201280T31 or Yeebo YB-TG7201280C69A-C-A1.
Casil backlight is 34V nominal, 32V to 38V at 25C, 20mA.
Yeebo backlight is 33V nominal, 32.4V to 38V at 25C, 20mA.
The TPS61161A can generate 17.6mA to 18.7mA with a 1 Ohm 1% sense resistor.

MIPI lines
Max data rate is 2.1Gbps.

LDO6_1V8_DSI_AND_IO net can be from 1.65V to 3.3V.
LDO17_2V85_ANALOG net can be 2.5V to 3.3V.
CTP_ANALOG_VDD net can be 2.8V to 3.6V.